

Strategic Order Splitting and the Demand / Supply of Liquidity[♦]

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Abstract

We identify and characterize order splitting strategies in an automated limit order market. We model the market conditions and order characteristics, which lead to the use of order splitting strategies. We find a positive correlation between price aggressiveness and the propensity to split an order, which implies, in a utility maximization framework, that traders trade-off one type of aggressiveness for another. We are able to identify two types of traders that use order splitting strategies for different purposes: the reduction of execution costs through split market orders, and voluntary supply of liquidity through split limit orders. Both order splitting strategies have the potential to reduce price volatility and smooth shocks to liquidity.

JEL classification: G10, G14, G15, C35.

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Strategic Order Splitting and the Demand / Supply of Liquidity

Market liquidity varies over time since the arrival of buyers and sellers to the market is not synchronized. Markets usually rely on designated market makers to bridge liquidity gaps and reduce the price volatility generated by the trading process. Since market liquidity is not perfect and traders cannot execute any order size at any time costlessly, they execute orders strategically. In this paper, we study the order splitting strategy, designed to minimize trading costs but also has the potential to reduce price volatility and smooth shocks to liquidity even in the absence of designated market makers.

Most studies tend to concentrate on one aspect of order submission while, in reality, this decision is rather complex and involves many dimensions that are determined simultaneously. The price aggressiveness of an order (the decision whether to submit a market order - demanding liquidity, or limit order - supplying liquidity) has been studied extensively. Time to execution and the different risks associated with it (non-execution, adverse selection and picking off risks) have also been studied recently. Even though order splitting is a common strategy designed to reduce trading costs, it did not receive much attention in the empirical literature.

Explicit costs of trade, such as commission fees, were in the center of attention in the past, but there has been a steady reduction in those costs over time. Empirical studies show that those costs are low relative to the implicit costs of trade. Four main types of implicit trading costs have been suggested in the literature: cost of execution,¹ costs/risks of non-execution, adverse selection and picking-off.² In a utility maximization framework, we can imagine a trader taking all costs/risks into account and choosing an order submission strategy that maximizes his utility. Recent examples of such models include Harris (1998), Parlour (1998), Foucault et al. (2005) and Hollifield et al. (2006). In practice researchers usually observe the trader's strategic choice, which

¹ A common measure for the cost of execution of market orders is the price impact. In general, we may think about such a cost for any order as the deviation of execution price from trader's reservation price. In practice, it is measured for market orders as the deviation of order execution price from the midquote.

² In the case of asymmetric information, uninformed traders face the adverse selection risk since they may trade with an informed trader. If new information arrives to the market, all limit orders in the book face the picking off risk since they may become mispriced and may be executed before the trader has an opportunity to cancel.

has simultaneous and usually opposite effects on execution costs/risks, and the end result is the outcome of an unobservable trade-off among them.

Pérol (1988) demonstrates the magnitude of trading costs and shows that ‘implementation shortfall’ significantly reduces the gains of a portfolio of the Value Line Fund: from about 20% on paper to 2.5% in practice. For large orders, the relative and absolute magnitude of those trading costs is much higher than for small orders and, as a result, they are almost always split into smaller parts. Chan and Lakonishok (1995) and Keim and Madhavan (1995, 1996) document order splitting activity of large institutional traders, while Bertsimas and Lo (1998) and many practitioners develop trading strategies to minimize the execution costs for such orders.³ In this paper, we try to fill a gap in the empirical literature. While Chan and Lakonishok (1995) and Engle et al. (2008) concentrate on a sample of split orders, we compare split to non-split orders and study the strategic decision to split.

Execution strategies associated with large orders affect both informed (as modeled by Easley and O’Hara, 1987) and liquidity traders (as modeled by Admati and Pfleiderer, 1988). The informed trader would like to hide his private information, which can be inferred from his order size, to maximize gains from trade; while the discretionary liquidity trader would like to hide his demand for liquidity to avoid front running and minimize the implicit cost of trade (see Keim and Madhavan, 1995, 1996). Regardless of the traders’ motivation for trade, large traders have an incentive to hide by splitting their orders to reduce implicit trading costs.

In the setting of an electronic limit order book without designated market makers, Bloomfield et al. (2005) find that a third motivation for trade may emerge – voluntary market making. In their experimental study, the market maker is more likely to be an informed trader since he does not face the adverse selection risk. In a symmetric-information setting with inventory risks/costs discussed in Stoll (1978), the role of liquidity supply is assumed by traders with a diversified portfolio. Therefore, we expect some traders to assume the role of liquidity suppliers even when they do not have superior information. However, the motivation for trade is no longer gains from information but rather the ability to extract a rent by supplying liquidity to the market.

³ Practitioners exert extensive effort to develop optimal order execution algorithms, in which splitting strategies play a dominant role. Algorithms of trading strategies and estimates of price impact are widespread in the investment management industry.

The liquidity supplier has an incentive to closely follow and monitor changes in market conditions, and may appear to be splitting limit orders.⁴

We use data from the Tel Aviv Stock Exchange (TASE), which is a pure limit order market without designated market makers or hidden orders. We do not run into problems that arise from order flow fragmentation since, most of the time, TASE is the only trading venue for the stocks in our sample.⁵ Another unique feature of the TASE is that all orders are routed through the same automated system and internal crossing of orders at the brokerage house is not allowed. This way, we are sure that the observed orders are all the orders submitted to the market. Odd lots are allowed on the TASE, and we are able to use this feature to our advantage to identify order splitting activity. In summary, we are able to avoid problems that arise from using small-size samples, aggregate data, and trade direction inferences from prices.

Splitting an order can be viewed as reducing order aggressiveness just as submitting a limit order is reducing price aggressiveness. We model both dimensions of order aggressiveness in a system of simultaneous equations, and allow for an unobserved effect that may drive both dimensions of aggressiveness.⁶ We find that the propensity to split an order and the propensity to submit a price aggressive (market) order are positively correlated. Our finding supports the existence of an unobserved trade-off between the two types of order aggressiveness. Our findings support the hypothesis that traders who split market orders minimize the implicit execution costs, while traders who split limit orders supply liquidity for a rent.

We find that traders who demand liquidity and submit split market orders reduce the total order price impact. We also find support for the conjecture that traders who supply liquidity and appear to split limit orders act as voluntary market makers. On average, they submit sequences of small but price improving orders, which add depth and narrow the spread. Splitting is a major component of activities such as high frequency and algorithmic trading, which serve both

⁴ The recent debate about High Frequency Trade in the media demonstrates that some trader profit from the voluntary supply of liquidity and make significant investments (fast computers, algorithms, fees to exchanges etc.) to be the first to offer liquidity.

⁵ Even for dual listed stocks in our sample, the trade on TASE overlaps the trade in US markets only during the last 45 minutes of the day.

⁶ The tradeoff between execution cost and risks is probably captured by other variables, such as the times to execution and cancellation, which we do not model here. In this sense, our model of the order splitting and price aggressiveness decisions is a reduced form of a more complex optimization problem. Incorporating those aspects will increase the dimensionality of the problem, complicate estimation and divert attention from the topic of this study.

demanders and suppliers of liquidity. Our findings suggest that those activities designed to maximize traders' profits, may have a smoothing effect on liquidity.

The paper is organized as follows: Section I presents a review of relevant literature; we describe the data and the TASE trading system in Section II; we define and characterize split orders in Sections III and IV; we present our results in Sections V, VI and VII. Section VIII concludes.

I. Literature and Hypotheses

Our paper is related to the literature on order submission strategies. The primary concern of many empirical and theoretical papers in this area is the study of price aggressiveness. Only a few studies incorporate the decision to split large orders even though, in practice, splitting is used extensively to reduce the cost of order execution. To form our hypotheses, we discuss the recent literature on order splitting and order size. To supplement our discussion, we also draw an analogy between splitting and price aggressiveness, and borrow some empirical predictions from the literature on price aggressiveness.

Below, we discuss three questions of interest in the study of order splitting: the incentives to split large market orders; the relation between price aggressiveness and splitting and a possible role of split limit orders.

A. Split market orders

We conjecture that traders who split market orders may be sophisticated/institutional traders who need to initiate or liquidate a position, but try to minimize price impact; or informed traders who try to maximize gains from trade.

By definition, noise/non-discretionary liquidity traders must trade the total exogenously determined quantity immediately, while discretionary liquidity traders may choose to split their orders over time (Admati and Pfleiderer, 1988) or across trading venues (Glosten, 1994) to minimize the implicit costs of trade. Vayanos (1999) studies the strategic behavior of large

uninformed discretionary liquidity traders, such as mutual funds and pension funds, in a dynamic model. He finds that large traders have a strong incentive to split their orders to minimize the price impact, but the speed of trade depends on traders' ability to hide their demand for liquidity. Vayanos (1999) predicts that large traders will split orders and trade slowly if their demand for liquidity is unknown to others, but they will trade faster and try to minimize front running losses if this demand is disclosed.⁷

Some theoretical models endogenize the order size decision for informed traders (see Kyle, 1985; Easley and O'Hara 1987; Admati and Pfleiderer, 1988; and Glosten, 1994). In a dynamic setting (Kyle, 1985), informed traders may submit a sequence of orders to reduce the rate of price adjustment to private information and increase their profits. Barclay and Warner (1993) are among the first to study the order size submitted by informed traders. They find supportive evidence to their stealth trading hypothesis that attributes medium size orders to informed traders.

Market orders provide certain and immediate execution at a high cost. On the other hand, limit orders reduce the price aggressiveness of the order and reduce this cost, but their execution is no longer fast or certain. Just as some patient traders choose limit over market orders, they may prefer to split large market orders to reduce execution cost at the expense of delayed execution and uncertainty about execution time and price.⁸ We conjecture that splitting a large market order reduces order aggressiveness just as submitting a limit order reduces price aggressiveness. Therefore, we borrow the following predictions for the effect of market conditions on splitting from the literature on price aggressiveness.

Book depth: The price aggressiveness decision in Parlour (1998) depends on the state of the limit order book and, in particular, book depth. If the buy-side depth is high, non-aggressive buy-side orders have low probability of execution and we will observe more price aggressive orders that "jump the queue". This prediction is exactly reversed for a buy-side order when the sell-side depth is high, and we expect fewer buy-side market orders. Applying the predictions in

⁷ Vayanos (1999) and others (Obizhaeva and Wang, 2005; Rostek and Weretka, 2007) go further than just endogenizing the strategic order size decision in a dynamic model. The order submission strategy and, in particular, order size is not only affected by market conditions (liquidity) but also has an effect on the market. In such models, market liquidity evolves endogenously and the execution strategies of large orders have a significant effect on market liquidity. This is beyond the scope of our paper.

⁸ For example, Engle and Ferstenberg (2007) model the trade-off between execution risk and cost for split orders in a mean variance framework.

Parlour (1998) to our setting will make a non-split market order the most aggressive strategy. While in Parlour (1998) the only way to increase aggressiveness is to submit a market order, in our setting the trader has more choice since he can both submit a market order and not-split his order to increase order aggressiveness.

Volatility: In general, the effect of volatility on order aggressiveness is not determined and may be either positive or negative depending on the magnitude of opposing effects.⁹ For example, a mechanistic effect of the volatility may increase the probability of execution, since high volatility of the price process increases the probability that any barrier (bid or ask prices) will be hit. Therefore, execution probability of limit orders increases and we expect price aggressiveness to decrease. A similar prediction of the effect of volatility is motivated by the risk of being picked off. In Foucault (1999) limit orders cannot be cancelled and for any given limit order, an increase in price volatility increases the risk of being picked off. Therefore, the price aggressiveness of limit orders decreases, the spread increases and market orders become expensive. The end result is fewer market orders and fewer price aggressive limit orders in the total order flow. On the other hand, Hasbrouck and Saar (2002) find the opposite effect – price aggressiveness increases when the volatility increases. Hasbrouck and Saar argue that the certainty effect of market orders can lead to a positive effect of volatility on price aggressiveness, since price uncertainty increases the dispersion of wealth outcome of a limit order. This interpretation relies on Cohen et al. (1981) that assumes that traders are risk averse.

Order arrival rate: The empirical implication of Foucault et al. (2005) is that high order arrival rate should decrease price aggressiveness since it increases the execution probability of all orders, including those that are not aggressive. **Bid-ask spread:** Another implication of Foucault et al. (2005) is that market orders are expensive when the spread is large and we will see more limit orders in the order flow when the spread is large. Consequently we expect to see more price aggressive orders when the order arrival rate is low and the spread is narrow. However, we do not see a direct link between the predictions for price aggressiveness and a prediction for order splitting.

Restricted order size: Moulton (2005) identifies cases in the foreign exchange market in which the order size is more likely to be restricted. She argues that at the end of the quarter traders

⁹ See Hasbrouck and Saar (2002) for a thorough discussion and investigation of all the possible effects.

are likely to submit the total (exogenous) order size, and finds that on those days they use price aggressive orders. Moulton (2005) is able to establish that traders' behavior is different when we expect their order size to be restricted, but she does not identify order splitting. Analogous to her findings for the end of the quarter, if the trader arrives to the market at the end of the day or at the end of the week and his order size is large, he is more likely to submit an aggressive order.

Asymmetric price impact is predicted by Kraus and Stoll (1972), Madhavan and Smidt (1991) and documented in Keim and Madhavan (1994, 1995) for institutional traders. They argue that prices react asymmetrically to buy- and sell-initiated trades, and the market reaction to buy-initiated trades is stronger than the reaction to sell-initiated trades. Therefore we expect buy-side orders to have longer execution times, which also imply that buy-side traders have an incentive to split their orders more than sell-side traders.

Intraday patterns: One implication of Admati and Pfleiderer (1988) is that time-of-the-day may have a significant effect on the order submission strategy. Other empirical studies also find patterns of the spread, volume, and volatility over the course of the trading day (e.g., Harris, 1996; McInish and Wood, 1990 among many others). Specifically for strategies of order splitting, Biais, et al. (1995) suggest that intra-day pattern could be the result of order splitting by strategic investors, who unwind their remaining exposure at the end of the day. For example, a mutual fund is evaluated at the closing price and therefore, tries to execute the remaining portion of orders by the end of the day. In a theoretical model, Obizhaeva and Wang (2005) show that large portions of the order are traded at the beginning and the end of the trading interval and smaller portions are traded in between, which implies more splitting in the middle of the trading day.

B. The relation between price aggressiveness and splitting

In this section we explore the relation between two dimensions of order aggressiveness - price and splitting. There are two extreme cases: no correlation or perfect correlation between those dimensions; both imply that order splitting should not affect the modeling and estimation of price aggressiveness. First, if the two decisions are uncorrelated, we can model them separately and the results will be unbiased. In this case it may be interesting to study order splitting, but this can be done separately for the rare cases in which one can actually observe the decision to split an order in the data. Second, if the two decisions are perfectly correlated, a similar conclusion applies.

If we have data on order splitting, we can simply refine our definition of order aggressiveness. In this case we can define one variable with the categories: market order (the most aggressive) then, say, split market, limit and split limit (the least aggressive order type). In this case, the data on order splitting allows us to define a more detailed response variable. Yet, if we don't have information about order splitting, we can just define a less detailed response variable for price aggressiveness without introducing systematic bias.

However, we conjecture that price aggressiveness and splitting are two distinct but related dimensions of order aggressiveness, and the correlation between the two is neither perfect nor zero. In a utility maximization framework, we can imagine a trader arriving to the market with an unobservable urgency to trade (impatience / demand for immediacy), and then choosing a strategy comprised of both aggressiveness dimensions. This trader may be trading off the costs and benefits of splitting and price aggressiveness, given his unobservable characteristics and the market conditions at the time of his arrival, to maximize utility from trade. Therefore, the correlation between order splitting and price aggressiveness may be significantly different from zero, +1 or -1, and the two decisions should be modeled in a simultaneous equations framework.

Keim and Madhavan (1995) study an ordered model of order aggressiveness, in which the aggressiveness of working orders (which are practically split orders) is ranked between market and limit. This could be an appropriate approach if the two aggressiveness dimensions could be reduced to one scale (i.e., if they are perfectly correlated). Engle et al. (2008) study the execution costs and risks of varying levels of order urgency, but they do that exclusively for a sample of split orders. Many other empirical studies of price aggressiveness do not have data on order splitting and simply ignore this dimension.¹⁰ One study that tries to take into account more than one dimension of order aggressiveness is Lo and Sapp (2007), even though they do not explicitly model order splitting. In a foreign exchange market setting, they model order size and price aggressiveness as two endogenous variables in a system of simultaneous equations and find that the submitted (endogenous) order size is negatively related to price aggressiveness.

Consistent with our conjecture that the price aggressiveness and order splitting are two distinct but related dimensions, we model them in a simultaneous equations framework, and empirically test for the sign and magnitude of the correlations between the two.

¹⁰ Some examples are Biais, Hillion Spatt (1995), Griffiths et al. (2000), and Ranaldo (2004).

C. Split limit orders

We conjecture that traders who appear to be splitting limit orders probably do not have a specific order size in mind and voluntarily supply liquidity for a rent (the bid-ask spread). Bloomfield, O'Hara and Saar (2005) find that voluntary supply of liquidity may emerge endogenously in a pure order driven market. In their experimental study, at the end of the trading session when the value of information is low, informed traders assume the role of voluntary market makers.

A comparison between split market and split limit orders helps motivate the split limit strategy and relate it to voluntary supply of liquidity. Splitting market orders reduces execution cost but increases non-execution risk. On the other hand splitting limit orders, which also increases non-execution risk, does not affect execution cost. Therefore, a cost reduction argument cannot explain the prevalence of split limit orders observed in our data. A possible motivation for splitting limit orders is active monitoring of changes in market conditions to reduce inventory risk and the risk of being picked-off. Those risks are the primary concerns of a market maker, and we conjecture that traders who split limit orders do so to profit from voluntary supply of liquidity.¹¹ Yet, unlike designated market makers, voluntary liquidity suppliers have no obligation to supply liquidity when the 'going gets tough'. Therefore, we do not expect them to make a market in extreme market conditions (for example, when prices are volatile).

In our view, splitting a limit order is not a decision about the optimal order execution strategy discussed above (Sections I.A and I.B). A model of split limit orders captures the stock and market characteristics that attract voluntary suppliers of liquidity, while the model of order aggressiveness (price and splitting) captures the strategic behavior of traders who are not voluntary suppliers of liquidity. Since those are different questions we model them separately.

Lastly, some markets allow the submission of hidden limit orders,¹² which may be considered as a special case of limit order splitting. In a recent empirical work on hidden orders,

¹¹ We do not observe the trader's risk tolerance or whether he has asymmetric information. However, all voluntary liquidity suppliers, whether they are informed or not, are exposed to inventory risk and the risk of being picked off.

¹² Hidden / iceberg orders are limit orders in which only a certain portion of the quantity is publicly disclosed. As soon as one part of the order is executed the next part is disclosed, but the order loses its time priority and gets to the end of the queue at the limit price.

Bessembinder, Panayides and Venkataraman (2007) study the joint decision of order price, size and exposure on Euronext Paris. On Euronext hidden orders are allowed, which makes it possible to expose only part of the demand for liquidity. Bessembinder et al. (2007) use a system of simultaneous equations to capture the relations between the total order size, the exposed proportion of the order, price aggressiveness and other cross sectional and intraday control variables. D'Hondt, De Winne and Francios-Heude (2005) is another study of hidden orders,¹³ which shows that even when hidden orders are allowed some traders may choose to split their orders. This finding suggests that hidden and split orders are not necessarily substitutes but rather complimentary strategies. The main difference between the two strategies is the degree of market monitoring and, consequently, the flexibility and speed of traders' response to changes in those conditions. By submitting a sequence of limit orders the voluntary market maker may keep his place at the head of the queue, while every new part of the limit order moves automatically to the end of the queue at the limit price. More importantly, a hidden limit order is still exposed to the risk of being picked off, while a split limit order does not expose the voluntary supplier of liquidity to this risk and allows him to quit at any point in time.

II. Data and sample

A. TASE trading system

The Tel Aviv Stock Exchange (TASE) is an electronic limit order book without designated market makers. Trading starts with a call auction at 9:45; then a continuous trading phase is held between 9:45 and 17:15, followed by a closing trading phase between 17:15 and 17:30. All unexecuted orders are cancelled at the end of every trading day, and the next day starts with an empty limit order book. We analyze order submission strategies in the continuous trading phase. All TASE traders observe the best three prices and quantities on each side of the limit order book. They may submit either market or limit orders, which are executed subject to price and time priority rules. Some of the stocks in our sample are dual-listed on TASE and US markets (most on NASDAQ). However due to time difference, simultaneous trading of those stocks on TASE and

¹³ Some examples of other hidden orders studies are Aitken, Berkman and Mak (2001) – Australian Stock Exchange; and Tuttle (2006) – NASDAQ.

on US venues takes place only during the last hour of TASE trading session. On TASE, there are four different tick size regimes defined by the stock price. The lowest tick size is NIS 0.001 (1 \$US = 4.5 New Israeli Shekel for sample time period) for prices below NIS 10, and NIS 1 for prices above NIS 1,000. In the continuous trading phase, traders may submit market or limit orders that are priced in multiples of the tick size and exceed a minimum order size of about NIS 2,000. There is no round lot restriction on TASE.

B. Sample

Our sample consists of the 40 most liquid stocks on TASE¹⁴ traded between July 1, 2006 and December 31, 2006.¹⁵ Table 2 reports cross-sectional descriptive statistics for the sample stocks. The average daily trading volume is NIS 18,422 million, but there is a lot of variation in the trading volume from a minimum of NIS 3,477 million to NIS 89,838 million. The average percentage spread is 0.68, but it is as low as 0.06% for the most liquid stock.

TASE provided the data on order submissions, cancellations, continuously-calculated limit order book status, and trades for all stocks. Each order in our sample is time stamped and matched with the book status prior to its submission and with the subsequent trades and/or cancellation. We use orders submitted during the continuous phase, starting 15 minutes after the beginning of trade in that phase (10:00) and ending at 17:15.¹⁶

In the continuous trading phase traders may submit two types of orders, market or limit, but pure market orders are very rare (less than 0.5%). Instead, they submit marketable limit orders, i.e. limit buy orders priced above the best ask or limit sell orders priced below the best bid. For the purposes of our analyses we treat these orders as if they were market orders, since the trader's main intent is clearly to obtain immediacy of execution. About one third of the observations in our sample can be categorized as market or marketable limit orders.

¹⁴ We chose the stocks with the largest trading volume on TASE in the three months prior to the sample period.

¹⁵ The sample period is relatively long and we get a very large data set for 40 liquid stocks on TASE. However, the fraction of split orders that we are able to identify is very small, and therefore, in some analyses we end up with a relatively small sub-sample. We will elaborate on the definition of sub-samples later on.

¹⁶ We define some of the explanatory variables as functions of the order flow or transaction rate 15 minutes prior to the submission of the order. Therefore, we only use orders submitted 15 minutes after the continuous trading phase begins.

All unexecuted orders on the TASE are automatically cancelled at the end of the trading day, yet about 37% of the orders are actively cancelled by the traders before the end of the trading day.

III. Classification of Split Orders

Strategic splitting of large orders is common practice. In our data, we observe sequences of orders for the same stock, of the same size and price, submitted on the same side (buy / sell) within a very short time interval (no more than one minute apart).¹⁷ We present an example of a split order in Table 3. In our view, it is very unlikely to observe such sequences by chance, and we use the combination of these properties to statistically identify split orders. We classify such sequences of orders as split orders, while orders that do not follow this pattern are classified as regular.¹⁸ Most sequences in our sample are rather small (a median of 3 orders per sequence) but there is a non-negligible probability of observing a much longer sequence.

We use a rather restrictive rule to identify sequences of split orders, and we are certain that there may be many other splitting strategies.¹⁹ However, in the following discussion of descriptive statistics for split orders, we present evidence supporting the assertion that the sequences we are able to identify are indeed split orders. Therefore, we are confident that our classification of split orders is rather accurate even though that of ‘regular’ orders may be noisy.

In Table 4 we present the distribution of sequence size and the probability of observing a round lot.²⁰ It is well documented that traders tend to use round numbers for prices and quantities (Moulton, 2005). Therefore, the order size is more likely to be a round number if it is generated

¹⁷ In other studies, such as Chan and Lakonishok (1995) the identity of the trader is known and the study follows the same manager for several days, assuming that all orders submitted during that period constitute one large order. That will not be a reasonable identification strategy in our study since we do not know the identity of the trader. We also believe that nowadays, in the age of high frequency and algorithmic trading (see Hendershott, Jones and Menkveld, 2007), it is more likely to observe much faster order sequences and shorter execution horizons.

¹⁸ All sequences that are shorter than 3 orders are considered as regular orders to account for the possibility of a pure coincidence. We repeat all the analyses in the paper for sequences of 2 orders or more and find similar results. We also exclude rare sequences of 61 orders or more, which we consider to be extreme and possibly erroneous observations.

¹⁹ Bertsimas and Lo (1998) develop trading strategies to minimize the expected execution costs for large orders. They find that the naive splitting strategy (equal parts) may be optimal under certain conditions and mention that in practice it is not uncommon.

²⁰ We will use the term round lot in this context to describe an order quantity that is a multiple of 10.

by a random grouping of orders into a sequence. Moreover, if the grouping is indeed a coincidence we expect the frequency of round lots to increase with the size of the sequence. In Panel A we compare the probability of round lots in the sub-samples of regular and split orders. The odds of a round lot in the sub-sample of split orders are about 0.60 versus 1.16 in the sub-sample of regular orders. In our view, this is evidence that the orders in a sequence are more likely to be parts of one big order, in which the total order size could be a round lot. In Panel B we show that the probability of a round lot in the sub-sample of split orders decreases as the size of the sequence increases. The odds of a round lot decrease from 0.69 for sequences of 3 orders to 0.22 for sequences of over 50 orders. Had our classification been an arbitrary grouping of unrelated orders from different traders, we should have observed the opposite.²¹

Most sequences of orders that define a split order are exclusively limit or exclusively market (marketable limit) orders. Only for 15% of the sequences we observe a combination of limit and market orders. We suspect that those sequences are the result of price changes between the time of order submission and the time of order arrival to the market.²² Had our identification strategy been an arbitrary grouping of orders into sequences, we should have seen a much higher percentage of mixed sequences. This, again, supports our statistical rule for classifying split orders.

The empirical and theoretical literature generates two main predictions for the size of split orders. Traders split orders that are large and tend to submit orders that are of medium or small size (Keim and Madhavan, 1995, 1996; Barclay and Warner, 1993). In Table 5 we present order size statistics consistent with those predictions. We use the quintiles of order size in the sub-sample of regular orders to determine thresholds and order size categories. Then, we turn to the sub-sample of split orders and classify every split order to a size category based on the thresholds. In Table 5, we compare the percentages of regular and split orders in each size category. Since we use size quintiles of regular orders to construct order size categories, we observe 20% of the regular orders

²¹ This result is even more pronounced for the less restrictive, conventional definition of a round lot (a multiple of 100).

²² For example, a trader might submit a sequence of marketable buy limit orders priced at the best ask, but another buy order may come in between, be executed against the best ask and consume all the liquidity offered at that price. As a consequence, the part of the original sequence of marketable limit orders will be classified as limit.

in every category but the percentages of split orders in those categories significantly deviate from 20%.²³

For a split order (sequence of orders) we consider two different size variables. One is the total size of the sequence (order size) and the other is the size of a single order in this sequence (part size). For the (total) order size, we observe a monotonic increase in the percentage of split orders as we move from the low to high order size categories. Only 0.2% of the split orders fall into the lowest size category and about 55% fall into the highest size category (Panel A). This observation is consistent with the benefits associated with splitting a large order, but it may simply be the result of the way we identify split orders. To rule out the possibility of arbitrary grouping of orders into sequences we also present the distribution of part size.

The distribution of part size of a split order significantly deviates from the uniform distribution and, therefore, it is different from the order size distribution in the regular sub-sample. There is a significant increase in the percentage of split market orders in the medium order size categories (Panel C: 25.42% for the second and 23.14% for the third size category) and a significant decrease in the percentage of split market orders in the higher order size categories (about 18%). This finding is consistent with the expectations that traders who split large market orders will hide their demand for liquidity in medium to small order size categories.²⁴ This finding is also consistent with Keim and Madhavan (1995, 1996).

For split limit orders the picture is somewhat different. There is a significant increase in the percentage of split limit orders in the lowest order size categories (Panel B: 22.61% for the first and 22.83% for the second size category) and a significant decrease in the percentage of split limit orders in the higher order size categories (about 18%). This is the first indication of the different roles of split market and split limit orders. While splitting market orders is designed to hide the demand for liquidity, traders who appear to be splitting limit orders probably do not have a specific order size in mind and need to monitor market conditions rather than hide.

²³ The rationale of this analysis is that of the q-q-plot. If two distributions are similar then we expect the same percentage of orders in every group, regardless of the sub-sample used to generate group thresholds.

²⁴ Even though we have the data to identify sequences of the same order size, this information is not available for traders who can observe only the last transaction and the three best quotes, which aggregate all orders at the same price. In fact, submitting a fast sequence of orders of the same size and price may be a good hiding strategy. The execution of such sequences is not easily detected, since a human trader will probably not notice a change in the last transaction if orders of the same size and price are executed seconds apart.

In summary, the characteristics of the distribution of round lots, price aggressiveness and order size in the sub-sample of split orders support our approach to the identification of split orders. Even though we applied a statistical rule to identify those orders, it is very unlikely that we identified sequences that are generated by chance and, at the same time, their characteristics corroborate the predictions for split orders.

IV. Characteristics of Split Orders

In this section, we study the order submission strategy, order execution and implicit execution costs of split and regular orders. Price aggressiveness (market/limit) and order side (buy/sell) distributions in the samples of split and regular orders are presented in Table 6. Regular orders are, on average, less aggressive than split orders (34.40% regular market orders versus 53.41% split market orders). This result may indicate that traders trade-off price aggressiveness and splitting. Traders who split reduce order aggressiveness since they reduce the submitted order size and, at the same time, they tend to submit market orders which increase price aggressiveness.

Buy-side traders tend to split orders more often than sell-side traders (58.88% buy-side split orders versus 41.12% sell-side split orders). This result is consistent with the asymmetric price response prediction in Kraus and Stoll (1972), Madhavan and Smidt (1991) and the findings in Keim and Madhavan (1995, 1996) for institutional traders. If indeed prices react asymmetrically to buy- and sell-initiated trades, and the market reaction to buy-initiated trades is stronger than the reaction to sell-initiated trades, then one should expect buy-side orders to have longer execution times as shown by Keim and Madhavan. This also implies that buy-side traders have more incentive to split their orders than sell-side traders.

The time to active cancellation is presented in Table 9.²⁵ Split orders are actively cancelled more often than regular orders (Panel A: 42% of the split orders versus 37% of the regular orders are actively cancelled); split limit orders are actively cancelled more often than split market orders (Panel B: 78.96% of the split limit orders versus 10.45% of the split market orders are actively cancelled). In the sub-sample of actively cancelled orders, split orders are cancelled faster than

²⁵ We consider an order cancellation to be active if the trader cancels the order before the automatic cancellation of all open orders at the end of the trading day.

regular orders, which is mainly due to very fast cancellations of split limit orders. On average, split limit orders are actively cancelled within 6:32 minutes while regular limit orders are actively cancelled within 29:17 minutes. This result is consistent with the conjecture that split limit orders are submitted by voluntary suppliers of liquidity who closely monitor the market.²⁶

One implication of Admati and Pfleiderer (1988) is that time-of-the-day may have a significant effect on the order submission strategy. Other empirical studies also show patterns of the spread, volume, and volatility over the course of the trading day (e.g., Harris, 1996; McInish and Wood, 1990, among many others). We find the familiar inverse J-shape for the intra-day distribution of trading volume. In Table 11 (Panel A), we show that 24% of the trading volume of regular orders (regular order volume at the beginning of the day to total daily regular volume) is traded at the beginning of the day. This volume goes down during mid day and increases again to 27% at the end of the day. For split orders, we also find a J-shaped pattern: 17% of the volume is traded in the beginning of the day, it goes down during mid day, and increases again to 37% at the end of the day. However, when we look at the intra-day pattern of split to total order volume, we see a monotonic increase during the day, with the highest percentage of split orders submitted at the end of the trading day (0.94% in the beginning versus 2.08% at the end of the day).

In Table 12, we sort the 40 stocks in our sample by average daily trading volume and form five ascending categories with eight stocks in every category. Then we calculate the relative frequency of split and regular orders for each category. In Panel A we observe a monotonic increase of the percentage of both types of orders as we move from the low to high volume categories. The percentage of split orders increases from about 9% to 37%, and the percentage of regular orders increases from about 7% to 42%. This increasing pattern is not surprising since we expect to see more orders for higher volume stocks.

The interesting result is the monotonic increase of the odds ratio of split to regular market orders in Panel C. This suggests that traders are more likely to split market orders when they trade high volume stocks, probably trying to reduce the price impact when the liquidity is high and the

²⁶ One strategy adopted by High Frequency Traders is very fast cancellations of orders designed to probe the market rather than to be executed. Even if the split limit orders we observe in our data are not designed to be executed, our conjecture that traders who submit those orders are voluntary suppliers of liquidity still holds. Either way, this highlights the differences between split limit and split market orders and supports our decision to model them separately.

non-execution risk is low. On the other hand there is an almost monotonic decrease in the odds ratio of split to regular limit orders in Panel B. This suggests that voluntary liquidity suppliers are more likely to trade low volume stocks, probably trying to extract a larger rent when liquidity is scarce.

In Table 13 we apply a similar analysis for the cross-sectional price-volatility. We sort the 40 stocks in our sample by average daily price volatility and form five ascending volatility categories with eight stocks in every category. Then we calculate the relative frequency of split and regular orders for each volatility category. We observe a monotonic decrease in the percentage of both types of orders as we move from the low to high volatility categories. The percentage of split orders decreases from about 48% to 8%, and the percentage of regular orders decreases from about 42% to 10% (Panel A). The interesting result is the high odds ratio for the least volatile stocks and low odds ratio for the most volatile stocks. This pattern suggests that more volatile stocks are also traded less frequently or that they have higher execution uncertainty. For traders who split market orders to reduce execution costs, splitting is riskier when prices are volatile. For traders who split limit orders to supply liquidity and extract a rent, splitting is also risky since the picking off risk increases.

It is generally accepted that market orders are aggressive relative to limit orders. Hasbrouck and Saar (2002) argue that fast order cancellations should also be considered as a type of order aggressiveness. Our analysis suggests that order splitting is another dimension of order aggressiveness, in which splitting an order, rather than submitting the total order size at once, represents a reduction in aggressiveness. At this stage, our univariate analyses suggest that traders trade-off the different types of aggressiveness. For example when they reduce aggressiveness by splitting, they either increase price aggressiveness or actively monitor and cancel their orders. They split large market orders and tend to hide in the medium order size range; they split market orders more frequently at the end of the trading day and more often for liquid and less volatile stocks. When traders voluntarily supply liquidity (split limit orders) they tend to submit smaller orders; they split limit orders more frequently at the end of the trading day and more often for less volatile and illiquid stocks.

V. Implicit Trading Costs

For split market orders we have a unique opportunity to observe both, the actual price impact of the split order and the counterfactual – the price impact of a hypothetical market order of same size that is not split. In this case, we can measure the effect of treatment (splitting) on the treated (split market orders), and quantify the effect of splitting on the execution costs. We use two variables to measure the implicit trading costs: the price impact and the average time to execution. The price impact of a split market order is measured as $(Midquote - Transaction Price)$ scaled by $Midquote$ at the time of first order submission, where $Midquote$ is the average of the best bid and ask prices.²⁷ We define a price impact of a hypothetical market order as $(Midquote - Hypothetical Execution Price)$ scaled by $Midquote$. The hypothetical execution price of a hypothetical market buy order is the prevailing best ask if the depth at the best ask is sufficiently high.²⁸

The price impact is analyzed in Table 10. As expected, we find that if the trader splits a large market order, the price impact of that order is significantly lower than the price impact of a similar market order that is not split (an average price impact of 17.20 basis points for a split market order versus 18.43 basis points for a hypothetical market order). Most of the difference in the price impact comes from the sub-sample of buy orders, which is consistent with the asymmetric price response story. This result implies that order splitting is indeed a successful strategy for reducing the price impact of market orders.

The average time to execution is a proxy for two other implicit trading costs – non execution and picking-off risks. Both costs increase with the expected time to execution. We present the average time in the book for split and regular orders in Table 7. On average, a sequence of split market orders stay in the book for 4:35 minutes while a regular market order stays in the book for 3:36 minutes (Panel B). This implies that the reduction in execution costs is traded off for an increase in non-execution risk. On the other hand, split limit orders stay in the book for 12:33 minutes, while regular limit orders stay in the book for 53:03 minutes on average. This is consistent with an attempt to reduce the picking off risk. The fact that a relatively small percentage of those orders is executed and the time to cancellation of those orders is short (as we show in

²⁷ This measure is commonly used by practitioners for regular market orders. For example see Almgren (2003).

²⁸ If the order cannot be completely filled at the best ask price, we assume that it will climb up the limit order book. If depth at the best three ask levels is not enough for complete execution, we assume that the remainder is executed at a price equal to the third ask level plus one tick.

Section III.A), supports the conjecture that those orders are generated by voluntary suppliers of liquidity who monitor them very closely.

In Table 8, we present the average time to execution for the sub-samples of completely and partially-executed orders. We find that, on average, split orders stay in the book for a shorter time period than regular orders (Panel A: 15:27 minutes for partially executed split orders versus 47:42 minutes for partially executed regular orders; 3:00 minutes for completely executed split orders versus 10:32 minutes for completely executed regular orders). This result is consistent with the second motivation for splitting large orders – reduction of the non-execution and picking-off risks.

VI. Order Splitting and Price Aggressiveness

In this section we focus on the decision to split market orders and its relation to price aggressiveness. We estimate a bi-variate probit model of price aggressiveness and splitting decisions, in which those decisions may be correlated. We define two latent endogenous variables: *price**, which represents the propensity to submit a price aggressive (market) order and *split**, which captures the propensity to split a market order. In general, we assume that splitting an order reduces aggressiveness, and we borrow the predictions of the theoretical literature for price aggressiveness to get similar implications for splitting market orders. The bi-variate probit specification allows for different effects of the explanatory variables on each dimension of order aggressiveness and for a test of the correlation hypothesis presented in Section I.B.²⁹

Let,

$$\begin{aligned} price^* &= \beta'x + \varepsilon \\ split^* &= \gamma'z + \eta \end{aligned}$$

where³⁰

$$\begin{bmatrix} \varepsilon \\ \eta \end{bmatrix} \sim N(0, R), \quad R = \begin{bmatrix} 1 & \rho \\ & 1 \end{bmatrix}$$

²⁹ Note that this modeling approach does not restrict one decision to be made before the other, rather allows for simultaneous consideration of both decisions. This structure allows for a more general correlation structure than a nested model.

³⁰ As in any discrete choice model, we have to impose identifying restrictions on the variance covariance matrix ($\sigma_1 = \sigma_0 = 1$).

Both latent variables are unobservable. Instead, we observe the following binary response variables.

$$price = \begin{cases} 1 & \text{if } price^* \geq 0 \text{ or } MKT \\ 0 & \text{if } price^* < 0 \text{ or } LMT \end{cases}$$

$$split = \begin{cases} 1 & \text{if } split^* \geq 0 \text{ or } split \text{ MKT} \\ 0 & \text{if } split^* < 0 \text{ or } non-split \text{ MKT} \end{cases}$$

Estimation results are presented in Table 15. We measure the Marginal Effect (ME) of a continuous variable X as the effect of one standard deviation from the sample mean on the probability, setting all other variables to sample mean values. For a binary variable D, the calculations is applied to the difference between levels 1 and 0 of D, while all other variables are set to sample mean values. The effects presented in the table may seem low, but this is a direct implication of the low proportion of split orders that we are able to identify.³¹

The intra-day pattern is consistent with our findings in the univariate analysis that splitting is more frequent at the end of the trading day. This finding is consistent with sophisticated liquidity traders who split more, also trade closer to the end of the day. However, this is not consistent with the prediction in Obizhaeva and Wang (2005) where most split orders are expected in the middle of the trading interval. For price aggressiveness, the least aggressive orders are observed in the morning while price aggressive orders are more likely at the end of the trading day. The day-of-the-week effects exhibit a similar pattern: at the end of the trading week traders submit price aggressive orders more frequently but they split their orders to reduce overall order aggressiveness.

Traders who arrive to the market with large orders (*Order Size*) reduce order aggressiveness by submitting either a limit order or a split market order. The interaction terms between order size and the end of the day (*OrderSize x last hour*) or end of the week (*OrderSize x last day*) restriction exhibit two polarized results. On one hand, patient traders whose trading

³¹ To demonstrate this point, consider a very simple model for the probability of some event $Pr(Y=1)$, which is explained by one binary variable D. For simplicity let us assume that $Pr(D=1)=0.5$. If the observed probability of the event is relatively high $Pr(Y=1)=0.3$ and we assume that $Pr(Y=1|D=1)=0.36$ while $Pr(Y=1|D=0)=0.24$ (20% deviation from the mean), then the additive marginal effect of D will be relatively large: $Pr(Y=1|D=1) - Pr(Y=1|D=0) = 0.36 - 0.24 = 0.12$. However, if we assume that the probability of the event is relatively low $Pr(Y=1)=0.008$ and we keep the effect of D on Y as 20% deviation from the mean, the additive marginal effect will be much smaller $Pr(Y=1|D=1) - Pr(Y=1|D=0) = 0.0096 - 0.0064 = 0.0032$. Had we been able to identify more cases of order splitting in our sample, the marginal effects would have been much larger. Therefore, the magnitude of the marginal effects presented in tables 15-16 should be compared to the percentage of split orders identified in our sample - 0.78%.

horizon is relatively long submit limit orders rather than price aggressive market orders. We also see that traders do not split market orders in this case. However, this does not tell us the whole story, since we do not know if traders who do not split increase order aggressiveness and submit a market order or decrease aggressiveness and submit a limit order. Here, it is informative to look at the conditional marginal effect of the interaction terms on splitting. While the unconditional marginal effect of *OrderSize* x *last hour* ($ME \times 100 = -7.095$) is relevant for the whole population (traders who consider market or limit orders), the conditional marginal effect is relevant only for the sub-population of traders who consider market orders (either split or non-split). Traders who arrive to the market at the end of the day and consider submitting a market order choose to increase order aggressiveness further and do not split (conditional $ME \times 100 = -19.339$). This result is consistent with Moulton (2005), where impatient traders tend to increase aggressiveness when order size is restricted. We also find that buy-side traders (*Buy*) are more likely to submit less aggressive limit orders or split their market orders, which is consistent with the finding in Keim and Madhavan (1995) that the perceived price impact of large buy-side orders is higher than that of sell-side orders.

Consistent with the empirical prediction of Parlour (1998), buy side traders increase overall order aggressiveness and submit non-split market orders when the bid-side depth (*Same Side depth*) is high. However, when the ask-side depth (*Opposite Side depth*) is high they are more likely to submit price aggressive market orders and, at the same time, decrease overall order aggressiveness by splitting. This result is a good example of the additional options traders have when deciding on the order submission strategy. If we use the restrictive Parlour (1998) setting, switching from market to limit order strategy is the only way to decrease order aggressiveness. If we add splitting as another dimension of order aggressiveness, then switching from market to split market order also decreases overall order aggressiveness.

High order arrival rate (*Order Arrival Rate*) decreases the probability of a price aggressive market order and increases the probability of splitting. This is consistent with the empirical prediction of Foucault et al. (2005), where high order arrival rate decreases price aggressiveness as it increases the execution probability of all orders in the book. We also estimate the effect of execution rate (*Trade Volume*), and find that high execution rate increases the probability to submit a price aggressive market order and the probability to split. We do not have theoretical predictions

for this variable, but we can try to explain the partial increase in order aggressiveness. On the one hand, high execution rate in the past may predict high execution rate in the future so traders may decrease aggressiveness by splitting their orders. On the other hand, high past execution rate may reduce the liquidity offered in the book and push traders to increase price aggressiveness and secure order execution.

The probability to split an order decreases when the average order size in the market (*Lag Order Size*) is high. Splitting is most likely to be considered for relatively large orders and, therefore, the size of recent orders submitted to the market should have an effect on this decision. The common practice to characterize the order size as a percentage of average daily volume demonstrates the importance of this variable for the order splitting decision, and we argue that the size of recent orders is relevant for the decision to split and hide.

A large bid-ask spread (*Spread*) implies high transaction costs for price aggressive orders. Therefore, we expect this variable to have a negative effect on order aggressiveness. We find that a large spread decreases the probability of a price aggressive market order but it also decreases the probability to split. In this case, again, it is informative to inspect the conditional rather than the unconditional marginal effect of the spread on the probability to split. Overall, we see fewer split market orders ($ME \times 100 = -0.052$), but this implies that we may see either more regular market (increasing order aggressiveness) or more limit orders (decreasing order aggressiveness). The conditional marginal effect of the spread on the probability to split a market order is positive (conditional $ME \times 100 = 0.044$), which implies that traders who end up submitting a market order tend to reduce order aggressiveness by splitting that order.

High intraday price volatility (*IVolatility*) increases the probability of a price aggressive market order and the probability to split. As we discussed in Section I.A, the effect of volatility on order aggressiveness may be either positive or negative depending on the magnitude of opposing effects. We find an increase in price aggressiveness, which is consistent with the empirical results in Hasbrouck and Saar (2002). Yet, there is an offsetting effect that decreases overall order aggressiveness through an increase of the probability to split.

We also measure the effect of an interaction between the bid-ask spread and the intraday volatility (*IVolatility* $\times$ *Spread*). In extreme market conditions, when the liquidity is scarce and uncertainty is high, we expect this interaction term to be large. In this case the probability of a

price aggressive market order is low. Yet, traders who demand liquidity and consider submitting a market order anyway, are likely to further increase order aggressiveness by not splitting.

The average daily trading volume (*Volume*) has a negative effect on the probability to submit a price aggressive market order and on the probability to split. This result is inconsistent with our univariate tests, and indicates that it is less likely to observe split market orders for liquid stocks. Controlling for other effects, this may indicate that splitting is not necessary; if the average daily volume is high traders can afford to submit the total order rather than split and risk delayed execution. Next, the cross sectional volatility (*Volatility*) has an overall positive effect on order aggressiveness. If the volatility is high, traders are more likely to submit price aggressive market orders and less likely to split. This result is consistent with the findings of Hasbrouck and Saar (2002). Lastly, we see more price aggressive market orders but also more order splitting activity in stocks that are included in the TA25 index and dual listing stocks. We attribute the prevalence of split orders to the fact that those stocks are traded by many sophisticated traders, including institutions and arbitrageurs.

In Section I.B we argue that the decisions of price aggressiveness and splitting should be modeled in a simultaneous equations framework, if the correlation between them is significantly different from zero, +1 or -1. First, the correlation reported in Table 15 is positive and equals 0.861. Since the standard error is 0.003, it is easy to see that the correlation is indeed significantly different from zero, +1 or -1. The positive correlation implies that, on average, the propensity to split market orders is positively correlated with the propensity to submit price aggressive market orders.

The correlation result could be a mechanical effect due to the fact that we only consider split market orders in this model. However, the different effects of market conditions and order characteristics on order aggressiveness imply that there are two distinct scenarios. On the one hand, traders may trade-off the two dimensions of order aggressiveness. For example, if the trading volume is high traders increase price aggressiveness by submitting market orders but also reduce aggressiveness by splitting. On the other hand, they may go for one of two extremes: traders who arrive to the market with large orders at the end of the day are either patient and, therefore, submit limit orders that may not be executed, or they are impatient / aggressive and submit non-split market orders. Since this result technically implies that the slope coefficients for the two

dimensions of order aggressiveness vary, it calls for the bi-variate specification of order aggressiveness: price and splitting.

VII. Split Limit Orders - Voluntary Supply of Liquidity

A. Univariate

In the case of split limit orders, there is no simple way to quantify execution costs. In fact, we did not expect to observe split limit orders at all. We suspect that the motivation to split limit orders is different from the motivation to split market orders. For a limit order, it is impossible to observe the counterfactuals and we do not have a clean measure of the effect of treatment (splitting) on the treated (split limit orders). Therefore, we compare the probability to improve on the quoted price and the price aggressiveness of split versus non-split limit orders. This provides evidence that further support our conjecture that split limit orders are probably submitted by voluntary suppliers of liquidity.

Throughout the paper we treat market orders, which demand liquidity, as price aggressive and limit orders, which supply liquidity, as less aggressive. To compare split and regular limit orders we refine our definition of price aggressiveness, and further distinguish between price improving limit orders and the limit orders that do not improve on the quoted price. A buy limit order is classified as price improving if the limit price is above the best bid and as non-improving if the limit price is at or below the best bid. Our refined definition of price aggressiveness allows for further investigation of the incentives to submit split limit orders.

In Panel A of Table 14, we compare the probability of a price improving limit order in two sub-samples: split and regular limit. Consistent with the expected behavior of voluntary liquidity suppliers, traders who split limit orders ‘jump the queue’ more often than traders who do not split (38.10% versus 27.64%). A comparison of the magnitude of price improvement is presented in Panel B. We find a significantly lower price improvement by split limit relative to regular limit orders (an average price improvement of 1.78 ticks versus 2.05 ticks).

B. Multivariate

The results of our univariate analyses support the conjecture that traders who split market orders are most likely sophisticated / institutional traders, who need to initiate or liquidate a

position and try to minimize price impact. We also find support for the conjecture that traders who split limit orders may be sophisticated traders who voluntarily make a market and extract a rent (the bid-ask spread). To allow for differences in the two populations of traders, we estimate a bivariate probit model of the two splitting decisions. We define two latent endogenous variables: $split_1^*$, which captures the propensity to split a market order and $split_0^*$, which captures the propensity to split a limit order. This way, we can investigate whether we observe the two types of order splitting under different market conditions.

Let

$$\begin{cases} split_1^* = \gamma_1'z + \eta_1 \\ split_0^* = \gamma_0'z + \eta_0 \end{cases}$$

where

$$\begin{bmatrix} \eta_1 \\ \eta_0 \end{bmatrix} \sim N(0, R), \quad R = \begin{bmatrix} 1 & \rho_{10} \\ & 1 \end{bmatrix}$$

The latent variables are unobservable and we observe binary response variables as follows:

$$split_1 = \begin{cases} 1 & \text{if } split_1^* \geq 0 \text{ or } split \text{ MKT} \\ 0 & \text{if } split_1^* < 0 \text{ or non-split MKT} \end{cases}$$

and

$$split_0 = \begin{cases} 1 & \text{if } split_0^* \geq 0 \text{ or } split \text{ LMT} \\ 0 & \text{if } split_0^* < 0 \text{ or non-split LMT} \end{cases}$$

To characterize the motivation to split we use intraday and weekly fixed effects, market conditions, order and cross sectional characteristics as our independent variables. A detailed description of all variables is presented in Table 1. Our discussion of the results in Table 16 will mostly highlight the differences between split market and split limit orders since we already discussed split market orders in Section VI. For split limit orders, which we interpret as voluntary supply of liquidity, we do not have theoretical predictions and we mostly relate to the results of the Bloomfield et al. (2005) experimental study.

The effects of order and cross sectional characteristics on the probability to observe splitting of market orders (split MKT) or voluntary supply of liquidity (split LMT) are the same.

So are the intraday and day-of-the-week effects. The main differences are observed for the effects of market conditions.

Discretionary liquidity traders or informed traders who try to minimize price impact split market orders when the bid-ask spread (*Spread*) is narrow, the intraday volatility (*IVolatility*) is high and the interaction between the two (*IVolatility* x *Spread*) is low. In Section VI we argue that split market orders are more attractive if the spread is narrow, which decreases the cost of market orders; and if volatility is high, which increases the execution risk of limit orders. On the other hand, voluntary suppliers of liquidity are more likely to participate in trade and extract a bigger rent when the spread is wide. However, if price volatility is high the risk of being picked off increases, and the voluntary supplier of liquidity does not have to take that risk since he does not have to trade at all. When the spread is wide and the intra-day volatility is low, we are more likely to observe voluntary supply of liquidity and less likely to observe split market orders.

In the univariate analyses in Section VII.A, we show that voluntary suppliers of liquidity tend to make small improvement on the quoted price. If the order arrival rate (*Order Arrival Rate*) is high but the execution rate (*Trade Volume*) is low, they can profit from providing liquidity if they submit price-improving limit orders. In this case many orders arrive to the market, and the voluntary supplier of liquidity can profit if his price-improving limit order motivates traders on the other side of the market to submit market orders. As for book depth, the same and opposite side depth (*Same Side depth* and *Opposite Side depth*) have negative effects on the probability of voluntary supply of liquidity. This result is not consistent with the finding of Bloomfield et al. (2005), where informed traders submit limit orders and make a market even when the same side depth is high. A possible explanation for our result is that high book depth means that there is enough liquidity offered by other traders and voluntary market makers are not able to profit by offering additional liquidity.

Lastly, we see a difference between the two types of order splitting on days with high trading activity in the opening stage (*ZVolume*), a variable that proxies for the arrival of new information to the market and may indicate a high level of uncertainty about future prices. We find a negative effect of this variable on the probability to split limit orders and no significant effect on the probability to split market orders. While liquidity traders have to participate in trade and

informed traders can profit from their superior information on those days, voluntary market makers do not have an obligation to trade and may decide to stay away in periods of high uncertainty.

It is not surprising that some market conditions have similar effects on traders who split market orders and demand liquidity and also on voluntary suppliers of liquidity. For example, large liquidity traders may have to execute orders at the end of the day but they split to reduce execution costs, while voluntary market makers may supply liquidity at the end of the day since the demand for liquidity is high. Thus, we expect an increase in order splitting activity at the end of the day for both types of traders. Another interesting variable is *OrderSize* $\times$ *last hour*. A large discretionary liquidity trader may be restricted at the end of the day, as we argue in Section VI. If such a trader arrives to the market at the end of the day with a large order he is less likely to split. So is the voluntary market maker. Inventory considerations will probably lead the voluntary market maker to do one of the two: not trade if he has no inventory or not split if he has a large open inventory position.

C. Is there a difference between split limit and split market orders?

We conjecture that split limit orders represent a strategy of voluntary market makers, while split market orders are assumed to be an order submission strategy used by other types of traders: either informed or discretionary liquidity traders. We show the different effects of market conditions on the probability to observe each of the two strategies in Section VII.B. Now, we use a multivariate approach to test whether those differences justify the distinction between the two splitting strategies.

We estimate a restricted version of the bi-variate probit discussed in Section VII.B: a univariate probit in which each explanatory variable has the same effect on the probability to observe a split market order and also on the probability to observe a split limit order.³² This univariate probit is nested in the bi-variate probit and we use a likelihood ratio test to show that the bi-variate specification (unrestricted model) is significantly different from the univariate probit specification (restricted model). Since we impose 29 restrictions, we have

³² For the univariate probit we only report the log likelihood. Estimation results are not presented in the paper and they are available from the authors.

$$X^2 = -2 \log \left[\frac{L_R}{L_U} \right] \chi^2(df = 29) \quad X^2 = 2(186,404 - 164,885) = 43,038$$

$$\Pr[X^2 > 43,038] < 0.0001$$

Therefore, the unrestricted model is significantly different from the restricted one. As we demonstrate in Section VII.B, the differences between split market and split limit orders are not merely statistical.

VIII. Conclusions

The Tel Aviv Stock Exchange (TASE) data provides a unique opportunity to identify order splitting, which may be used to hide large order volume and reduce the price impact of large market orders. We find that the price impact of a hypothetical market order is significantly higher than the price impact of an equivalent split market order, suggesting that order splitting is a successful strategy to reduce implicit execution costs. Our analyses show that the propensity to split an order is positively correlated with price aggressiveness. This implies a tradeoff between the two types of order aggressiveness: when traders submit a price aggressive market order they tend to reduce the overall order aggressiveness by splitting.

We conjecture that splitting of limit orders represents voluntary supply of liquidity and traders use those orders to reduce inventory and picking off risks. The average time to execution and time to cancellation of split limit orders is lower than that of regular limit orders, which is consistent with an active monitoring role of splitting. Traders who split limit orders tend to make small improvements on the quoted price, but they tend not to participate in trade when volatility and uncertainty are high.

Our findings on splitting of market orders and what appears to be splitting of limit orders imply that these strategies should smooth liquidity. When large market orders are split they reduce the price impact and, since those traders do not expect immediate execution of a large quantity, the shock to market liquidity is relatively small. If traders who appear to split limit orders are indeed voluntary market makers, they actually increase the available liquidity by increasing the

depth and decreasing the spread. Yet, these positive effects on liquidity are not always present. If the traders' urgency to trade is high we will not observe split market orders. If uncertainty and volatility are high we will not observe split limit orders. Thus splitting, which has a major role in high frequency and algorithmic trading, has a positive effect on market liquidity in normal times, but it is not present when the market needs it most.

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Table 1: Description of Explanatory Variables

Variable Name	Description
<i>Intraday & weekly fixed effects</i>	
Daytime	A discrete variable that indicates the time of order submission during the day. daytime={1 if 9:45-11:15 ; 2 if 11:15 - 12:45; 3 if 12:45 - 14:15 ; 4 if 14:15 - 15:45; 5 if 15:45 - 17:15}
Weekday	A discrete variable that indicates the day of the week. Weekday={First Day if first day of the week; Last Day if last day of the week; Mid Week otherwise }
<i>Market conditions</i>	
Same Side Depth*	Log of depth at the best bid.
Opposite Side Depth*	Log of depth at the best ask.
Order Arrival Rate	Log of the order volume submitted in the fifteen minutes time interval prior to order submission.
Trade Volume	Log of the transaction volume in the fifteen minutes time interval prior to order submission.
Lag Order Size	Log of the average order size measured in the fifteen minutes time interval prior to order submission.
IVolatility	Intra-day volatility measured in the fifteen minutes time interval prior to order submission.
Spread	log of spread size just before order submission.
IVolatility×Spread	An interaction between intra-day volatility and the bid-ask spread.
<i>Order characteristics</i>	
Order Size	Log of submitted order size.
Order Size×last hour	An Interaction between log order size and the indicator of last one and half hours the trading day.
Order Size×last day	An interaction between log order size and the indicator of last day of the week.
Buy	Buy side order indicator.
Round	Round quoted price indicator.
<i>Daily & Cross sectional variables</i>	
Price	Log price for each stock.
Volume	Log daily volume for each stock.
ZVolume	Z-score of daily opening stage volume for each day and stock.
Volatility	Log daily volatility for each stock.
TA25	Tel-Aviv 25 index indicator.
Dual Listing	Dual trading stock indicator.
Tick Size	Minimum price change {0.001 if price <10NIS; 0.01 if 10NIS≤ price<100NIS; 0.1 if 100NIS≤ price<1000NIS; 1 if price >1,000NIS}

* Variable definitions are symmetric for buy and sell side orders.

Table 2: Tel-Aviv Stock Exchange (TASE) Summary Statistics

The sample consists of the 40 largest stocks of TASE and the sample period is July to December 2006 (127 trading days). The stocks are selected based on their total trading volume in the three months preceding the sample period (April to June, 2006). The table reports cross-sectional distributions of following variables: **Mkt Cap** is the average daily market capitalization; **Price** is the average daily closing price; **Spread** is the percentage spread; **Daily Turnover** is the average daily turnover; **No. of Transactions** is the average number of daily trades; **Daily Volume** is the average daily trading volume; **No. of Orders** is the daily average number of orders; and **Order Volume** is the average daily order volume.

	Mkt Cap (NIS Mil.)	Price (NIS)	Spread (%)	Daily Turnover (NIS 100,000)	Daily No. of Transaction	Daily Volume (NIS 1000)	Daily No. of Orders	Daily Order Volume (NIS 1000)
Mean	9,144	133.90	0.68	19,182	52,283	18,422,116	50,009	77,558,095
Std. dev	17,927	10.98	0.60	26,280	46,842	13,153,957	39,726	65,463,454
Min	1,025	115.09	0.06	1,156	3,857	3,477,215	644	14,519,037
Q1	2,849	124.65	0.26	5,491	24,980	10,338,002	26,758	41,994,049
Median	5,042	131.49	0.53	10,843	35,584	15,060,461	47,633	58,682,020
Q3	7,369	143.97	0.93	22,135	64,295	22,688,186	75,313	87,936,964
Max	111,440	155.79	3.43	122,267	215,538	89,838,376	102,813	422,286,761

Table 3: Example of a Split Order

Ticker NICE, Date 07/13/2006

Type	Side	Price	Quantity	Order Time
Market	Sell	11.33	173	14:12:56.61
Market	Sell	11.33	173	14:13:21.35
Market	Sell	11.33	173	14:13:38.09
Market	Sell	11.33	173	14:13:46.86
Market	Sell	11.33	173	14:13:51.36

Table 4: Sequence Size and Frequency of Round Lots

Table 4 Panel A reports the observed frequency and odds of round lots for split and regular orders. Panel B reports the distribution of sequence size for split orders and the odds of observing a round lot for each size category. Each split order is a sequence of orders for the same stock, of the same size and price, submitted on the same side (buy/sell) within an interval of one minute. Sequence size refers to the number of parts observed for each split order. A round lot is defined as order quantity that is a multiple of 10. The sample consists of 40 largest stocks of TASE. The sample consists of the 40 largest stocks of TASE and the sample period is July to December 2006 (127 trading days).

Panel A

Type	# of Obs	Proportion	% Round	Odds (Round/Non-round)
Split Orders	35,701	0.78%	37.51%	0.6003
Regular Orders	4,526,771	99.22%	53.66%	1.1580

Panel B

Sequence Size	# of Obs	Probability of the sequence	% Round	Odds (Round/Non-round)
3 orders	22,279	62.40%	40.78%	0.6886
4	6,867	19.23%	35.43%	0.5487
5	2,610	7.31%	31.00%	0.4493
6	1,322	3.70%	29.65%	0.4215
7	656	1.84%	28.96%	0.4077
8	422	1.18%	28.67%	0.4019
9	295	0.83%	28.47%	0.3980
10	176	0.49%	30.11%	0.4308
11-20	568	1.59%	23.59%	0.3087
21-30	211	0.59%	18.01%	0.2197
31-40	97	0.27%	15.46%	0.1829
41-50	115	0.32%	20.00%	0.2500
51-60	83	0.23%	18.07%	0.2206

Table 5: Order Size Frequency by Regular Order Size Quintile

Table 5 reports the number and percentage of split and regular orders in each size quintile. Each split order is a sequence of orders for the same stock, of the same size and price, submitted on the same side (buy/sell) within an interval of one minute. Part size refers to the size of each part in a sequence of orders. Order size in the sub-sample of regular orders is used to determine thresholds and order size quintiles. 1 represents the smallest/lowest and 5 the largest/highest quintile. The percentage of split orders that falls into each order size quintile is reported in column (b). The percentage of split orders that falls in each part size quintile is reported in column (c). We report the statistics for all orders in Panel A, for limit orders in Panel B and market orders in Panel C. The sample consists of the 40 largest stocks of TASE and the sample period is July to December 2006 (127 trading days).

Regular Order			Split Order				
Panel A: All Orders			(a)		(b)		(c)
Order Size Quintile	# of Obs	Order Size (%) of obs	# of Obs	Order Size (%) of obs	# of Obs	Part Size (%) of obs	P-value*
1	898,214	20%	72	0.20%	7,330	20.53%	0.00
2	898,214	20%	1,547	4.33%	8,904	24.94%	0.00
3	898,214	20%	3,972	11.13%	8,036	22.51%	0.00
4	898,214	20%	10,589	29.66%	5,705	15.98%	0.00
5	898,214	20%	19,521	54.68%	5,726	16.04%	0.00
Total	4,491,070		35,701		35,701		
Panel B: Limit Orders							
Order Size Quintile	# of Obs	Order Size (%) of obs	# of Obs	Order Size (%) of obs	# of Obs	Part Size (%) of obs	P-value*
1	589,186	20%	57	0.34%	3,761	22.61%	0.00
2	589,186	20%	1,057	6.35%	3,797	22.83%	0.00
3	589,186	20%	2,156	12.96%	3,391	20.39%	0.00
4	589,186	20%	4,118	24.76%	2,726	16.39%	0.00
5	589,186	20%	9,245	55.58%	2,957	17.78%	0.00
Total	2,945,930		16,633		16,633		
Panel C: Market Orders							
Order Size Quintile	# of Obs	Order Size (%) of obs	# of Obs	Order Size (%) of obs	# of Obs	Part Size (%) of obs	P-value*
1	309,028	20%	15	0.08%	2,841	14.90%	0.00
2	309,028	20%	462	2.42%	4,847	25.42%	0.00
3	309,028	20%	1,021	5.35%	4,412	23.14%	0.00
4	309,028	20%	7,095	37.21%	3,378	17.72%	0.00
5	309,028	20%	1,0475	54.93%	3,591	18.83%	0.00
Total	1,545,140		19,068		19,068		

* $H_0: (c) - (a) > 0$ for size quintile 1, 2 and 3 and $H_0: (a) - (c) < 0$ for size quintile 4 and 5 for Panel A and B.

$H_0: (c) - (a) > 0$ for size quintile 2 and 3 and $H_0: (a) - (c) < 0$ for size quintile 1, 4 and 5 for Panel C.

Table 6: Distribution of Order Aggressiveness and Side

Table 6 reports order type statistics. In Panel A we present the distribution of orders submitted to TASE. We define each split order as a sequence of orders for the same stock, of the same size and price, submitted on the same side (buy/sell) within an interval of one minute. Ratios of the different order volumes are reported in Panel B. The sample consists of the 40 largest stocks of TASE and the sample period is July to December 2006 (127 trading days).

Panel A: Frequencies

Order Type	Regular	Split	Total
LMT	2,945,930 65.60%	16,633 46.59%	2,962,563 65.45%
MKT	1,545,140 34.40%	19,068 53.41%	1,564,208 34.55%
Buy	2,331,656 51.92%	21,020 58.88%	2,352,676 51.97%
Sell	2,159,414 48.08%	14,681 41.12%	2,174,095 48.03%

Panel B: Volume

$\frac{\text{Total Split Order Vol}}{\text{Total Order Vol}}$	$\frac{\text{MKT Split Order Vol}}{\text{Total Split Order Vol}}$	$\frac{\text{MKT Regular Order Vol}}{\text{Total Regular Order Vol}}$	$\frac{\text{Buy Split Vol}}{\text{Total Split Vol}}$	$\frac{\text{Buy Regular Vol}}{\text{Total Buy Vol}}$
2.59%	53.06%	36.17%	54.74%	97.15%

Table 7: Order Lifetime Distribution

Table 7 reports the distribution of order lifetime for split and regular orders, by order type. Each split order is a sequence of orders for the same stock, of the same size and price, submitted on the same side (buy/sell) within an interval of one minute. We classify an order as executed if at least 90% of order quantity is filled; An order is classified as cancelled if less than 90% of order quantity is filled and it is actively cancelled by the trader before the end of the trading day; An order is classified as censored if less than 90% of order quantity is filled and it is automatically cancelled by the exchange at the end of the trading day. The lifetime of an executed split order is the time between the submission of the first order in a sequence and execution of the last order in the sequence. If the last order in a sequence is cancelled (censored) then the split order is classified as cancelled (censored), and the lifetime of this order is the time between the submission of the first order in the sequence and the first cancellation (censoring) of an order in the sequence. The statistics for the entire sample are reported in Panel A. The statistics for the sub-samples of limit and market orders are reported in Panel B. The sample consists of the 40 largest stocks of TASE and the sample period is July to December 2006 (127 trading days).

Panel A: All Orders

	Regular	Split
Mean	0:28:19.8	0:08:34.4
Std Dev	0:28:24.8	0:09:54.9
Min	0:00:30.0	0:00:59.9
Q1	0:02:47.2	0:02:50.1
Median	0:17:50.1	0:04:46.2
Q3	0:45:50.0	0:09:25.1
Max	2:00:09.1	0:50:46.9
% Executed	56.10%	56.57%
# Obs	2,519,634	20,197

Panel B: Limit versus Market Orders

	Limit Orders		Market Orders	
	Regular	Split	Regular	Split
Mean	0:53:03.3	0:12:33.2	0:03:36.2	0:04:35.6
Std Dev	0:19:21.5	0:12:01.9	0:02:34.1	0:04:40.6
Min	0:21:44.4	0:01:08.4	0:00:30.0	0:00:59.9
Q1	0:39:38.7	0:04:48.9	0:02:04.2	0:02:31.5
Median	0:45:50.0	0:08:22.7	0:02:47.2	0:03:16.0
Q3	1:04:14.7	0:15:21.5	0:04:35.4	0:04:46.2
Max	2:00:09.1	0:50:46.9	0:13:55.8	0:27:50.1
% Executed	41.77%	15.86%	58.23%	84.14%
# Obs	1,052,503	3, 204	1,467,131	16,993

*Lifetime measured in hh:mm:ss.00

Table 8: Time to Execution

Table 8 reports the average time to execution for split and regular orders. Each split order is a sequence of orders for the same stock, of the same size and price, submitted on the same side (buy/sell) within an interval of one minute. We classify an order as executed if at least 90% of order quantity is filled; An order is classified as partially executed if at least one share but less than 90% of order quantity is filled. An order is classified as cancelled if less than 90% of order quantity is filled and it is actively cancelled by the trader before the end of the trading day; An order is classified as censored if less than 90% of order quantity is filled and it is automatically cancelled by the exchange at the end of the trading day. The lifetime of an executed split order is the time between the submission of the first order in a sequence and execution of the last order in the sequence. If the order is partially executed it is either censored or cancelled, and its lifetime is the time between the submission of the first order in the sequence and the first cancellation or censoring of an order in the sequence. The statistics for the entire sample are reported in Panel A. The statistics for the sub-samples of limit and market orders are reported in Panel B. The sample consists of the 40 largest stocks of TASE and the sample period is July to December 2006 (127 trading days).

Panel A: All Orders

	Regular		Split	
	Completely Executed	Partially Executed	Completely Executed	Partially Executed
Mean	0:10:32.5	0:47:42.3	0:02:59.9	0:15:27.3
Std. dev	0:37:32.4	0:27:08.6	0:14:24.9	0:16:44.3
Min	0:00:00.0	0:09:39.2	0:00:00.0	0:00:45.1
Q1	0:00:00.0	0:25:36.5	0:00:24.6	0:05:43.9
Median	0:00:03.0	0:40:09.9	0:00:52.2	0:10:29.4
Q3	0:03:24.8	1:02:51.3	0:01:34.0	0:19:25.4
Max	7:24:32.5	2:18:00.4	6:28:15.1	1:43:19.2
#Obs	4,526,771	1,971,436	19,068	15,504

Panel B: Limit versus Market Order

	Limit Orders				Market Orders			
	Completely Executed		Partially Executed		Completely Executed		Partially Executed	
	Regular	Split	Regular	Split	Regular	Split	Regular	Split
Mean	0:23:44.4	0:08:12.8	1:06:21.8	0:14:35.6	0:01:04.4	0:02:00.9	0:29:02.8	0:16:20.2
Std. dev	0:54:15.9	0:26:24.4	0:24:54.4	0:17:14.2	0:09:38.9	0:10:27.6	0:12:33.8	0:16:23.4
Min	0:00:00.0	0:00:00.6	0:20:29.0	0:00:45.1	0:00:00.0	0:00:00.0	0:09:39.2	0:01:01.8
Q1	0:01:00.6	0:00:58.2	0:48:18.7	0:04:14.2	0:00:00.0	0:00:21.2	0:20:52.2	0:07:19.0
Median	0:04:10.1	0:01:49.0	1:00:54.8	0:08:41.4	0:00:00.0	0:00:46.6	0:25:53.3	0:12:55.9
Q3	0:17:15.3	0:04:32.8	1:20:34.8	0:15:53.1	0:00:00.0	0:01:19.8	0:33:25.0	0:19:50.8
Max	7:24:32.5	6:28:15.1	2:18:00.4	1:27:44.3	7:20:13.1	6:18:17.7	1:16:05.6	1:43:19.2
#Obs	3,059,640	2,075	1,893,427	13, 429	1,467,131	16,993	78,009	2,075

*Time to execution is measured in hh:mm:ss

Table 9: Time to Active Cancellation

Table 9 reports the probability of active cancellation within a time interval. Each split order is a sequence of orders for the same stock, of the same size and price, submitted on the same side (buy/sell) within an interval of one minute. We classify an order as actively cancelled if less than 90% of order quantity is filled and it is actively cancelled by the trader before the end of the trading day. The lifetime of an actively cancelled split order is the time between the submission of the first order in the sequence and the first cancellation of an order in the sequence. The statistics for the entire sample are reported in Panel A. The statistics for the sub-samples of limit and market orders are reported in Panel B. The sample consists of the 40 largest stocks of TASE and the sample period is July to December 2006 (127 trading days).

Panel A: All Orders

Cancelled Within	Regular	Split
10 sec	10.84%	42.46%
5 min	56.28%	89.10%
10 min	67.72%	93.61%
30 min	82.27%	97.28%
60 min	88.78%	98.48%
# of Obs	1,662,756	15,126
% Active Cancellations	37.02%	42.37%
Mean Time to Active Cancellation*	00:28:34.56	00:07:26.59

Panel B: Limit versus Market

	Limit Orders		Market Orders	
Cancelled Within	Regular	Split	Regular	Split
10 sec	10.46%	47.84%	19.47%	7.18%
5 min	55.55%	90.76%	72.34%	78.16%
10 min	67.11%	94.49%	80.97%	87.85%
30 min	81.88%	97.53%	90.96%	95.68%
60 min	88.51%	98.64%	94.64%	97.39%
# of Obs	1,590,380	13,134	72,376	1,992
% Active Cancellations	53.99%	78.96%	4.68%	10.45%
Mean Time to Active Cancellation*	00:29:17.43	00:06:32.75	00:15:00.99	00:09:52.60

* Mean time to active cancellation is measured in hh:mm:ss

Table 10: The Price Impact of a Split Market Order

Table 10 compares the price impact of an actual split market order and an hypothetical market order. Each split order is a sequence of orders for the same stock, of the same size and price, submitted on the same side (buy/sell) within an interval of one minute. The total quantity of a sequence is measured as the sum of its parts. The price impact of a split market order is defined as $(\text{Midquote} - \text{Transaction price})/\text{Midquote}$ at the time the first order in the sequence is submitted. For every split market order we define the price impact of an hypothetical market order as $(\text{Midquote} - \text{Hypothetical Execution Price})/\text{Midquote}$, where the Hypothetical Execution Price is the execution price of the split market order had it been submitted as a regular market order. For an hypothetical market buy order, if the total quantity cannot be filled at the best ask price, we assume that it will climb up the book. All the statistics are calculated for each stock and then averaged across stocks, weighted by the number of orders for the stock. The sample consists of the 40 largest stocks of TASE and the sample period is July to December 2006 (127 trading days). All statistics are in basis points (bp).

	Hypothetical Market Order			Split Market Order		
	All	Buy	Sell	All	Buy	Sell
Mean	18.43 bp	19.24 bp	17.61 bp	17.20 bp*	17.78 bp*	16.63 bp
Std. dev	9.92	12.02	7.31	8.49	9.72	7.14
Min	6.80	6.93	6.80	4.72	4.72	5.48
Q1	11.58	11.37	12.86	11.29	11.19	12.65
Median	16.04	15.27	16.43	15.91	15.11	16.01
Q3	22.67	23.39	22.65	20.49	23.42	19.95
Max	62.91	62.91	40.61	46.86	46.86	37.50

* Differences between the mean price impacts of hypothetical and split market orders are significant at 1%

Table 11: Intra-Day Pattern of Order Volume

Table 11 reports the intra-day pattern of order volume for split and regular orders. Each split order is a sequence of orders for the same stock, of the same size and price, submitted on the same side (buy/sell) within an interval of one minute. The continuous trading phase (9:45am to 5:15pm) is divided into 5 equal 90-minute intervals. We report the statistics for all orders in Panel A, for limit orders in Panel B and market orders in Panel C. The sample consists of the 40 largest stocks of TASE and the sample period is July to December 2006 (127 trading days).

Panel A: All Orders

Daytime Interval	Regular Daytime Vol	Split Daytime Vol	Split Daytime Vol
	Total Regular Daily Vol	Total Split Daily Vol	Total Daytime Vol
9:45-11:15	23.85%	17.49%	0.94%
11:15-12:45	17.97%	14.31%	1.22%
12:45-2:15	15.27%	14.55%	1.38%
2:15-3:45	15.96%	16.50%	1.48%
3:45-5:15	26.95%	37.15%	2.08%

Panel B: Limit Orders

9:45-11:15	26.04%	20.22%	0.69%
11:15-12:45	17.65%	14.15%	0.88%
12:45-2:15	14.64%	11.87%	0.91%
2:15-3:45	15.37%	12.87%	0.91%
3:45-5:15	26.30%	40.89%	1.59%

Panel C: Market Orders

9:45-11:15	19.21%	15.88%	1.43%
11:15-12:45	18.66%	14.74%	1.63%
12:45-2:15	16.74%	15.64%	1.97%
2:15-3:45	17.32%	18.81%	2.30%
3:45-5:15	28.06%	34.93%	2.68%

Table 12: Order Frequency by Trade Volume Quintile

Table 12 reports the distribution of split and regular orders by liquidity quintiles. Each split order is a sequence of orders for the same stock, of the same size and price, submitted on the same side (buy/sell) within an interval of one minute. Liquidity is measured by the average daily trading volume and use it to rank and classify stocks into five liquidity quintiles. 1 represents the smallest/lowest and 5 the largest/highest quintile. We report the statistics for all orders in Panel A, for limit orders in Panel B and market orders in Panel C. The sample consists of the 40 largest stocks of TASE and the sample period is July to December 2006 (127 trading days).

Panel A: All Orders

Volume Quintile	# of Split Orders	# of Regular Orders	Odds ratio (Split/Regular)
	Total # of Split Orders	Total # of Regular Orders	
1	9.05%	7.16%	1.264
2	9.71%	11.66%	0.833
3	19.04%	15.78%	1.207
4	24.71%	23.51%	1.051
5	37.49%	41.90%	0.895

Panel B: Limit Orders

1	7.39%	4.58%	1.614
2	7.56%	8.07%	0.936
3	11.11%	13.16%	0.844
4	15.27%	18.36%	0.832
5	58.67%	55.83%	1.051

Panel C: Market Orders

1	2.03%	3.28%	0.620
2	4.44%	6.23%	0.713
3	8.85%	10.46%	0.846
4	15.03%	17.92%	0.839
5	69.65%	62.12%	1.121

Table 13: Order Frequency by Trade Volatility Quintile

Table 13 reports the distribution of split and regular orders by volatility quintiles. Each split order is a sequence of orders for the same stock, of the same size and price, submitted on the same side (buy/sell) within an interval of one minute. Volatility is measured as the average of price range, scaled by the midpoint of the range. We calculate it for every stock, as a daily average and use it to rank and classify stocks into five volatility quintiles. 1 represents the smallest/lowest and 5 the largest/highest quintile. We report the statistics for all orders in Panel A, for limit orders in Panel B and market orders in Panel C. The sample consists of the 40 largest stocks of TASE and the sample period is July to December 2006 (127 trading days).

Panel A: All Orders

Volatility Quintile	# of Split Orders	# of Regular Orders	Odds ratio (Split/Regular)
	Total # of Split Orders	Total # of Regular Orders	
1	47.80%	42.09%	1.213
2	17.23%	20.73%	1.106
3	20.59%	18.37%	0.881
4	6.51%	8.64%	1.146
5	7.88%	10.17%	0.487

Panel B: Limit Orders

1	50.50%	39.44%	1.319
2	15.12%	19.50%	1.046
3	13.49%	17.56%	0.766
4	12.03%	11.72%	1.017
5	8.86%	11.78%	0.588

Panel C: Market Orders

1	33.92%	28.84%	1.052
2	32.98%	33.26%	0.993
3	16.68%	18.28%	1.073
4	10.19%	10.18%	1.413
5	6.22%	9.45%	0.414

Table 14: Characterizing Split Limit Orders - Probability and Size of Price Improvement

Table 14 reports descriptive statistics for split and regular limit orders. Each split order is a sequence of orders for the same stock, of the same size and price, submitted on the same side (buy/sell) within an interval of one minute. Panel A reports the percentage of price improving limit orders and Panel B reports the size of price improvement (in ticks) for split and regular limit orders. A buy (sell) limit order is classified as price improving if the limit price is above (below) the best bid (ask), and it is classified as non-improving if the limit price is at the best bid (ask) or below (above). The sample consists of the 40 largest stocks of TASE and the sample period is July to December 2006 (127 trading days).

Panel A

	Split LMT	Regular LMT	Odds ratio (Split/Regular)
# of Price Improving LMT Total # of LMT Orders	38.10%	27.64%	1.3784
	# of Obs	Chi-square (df = 1)	P-Value
	90,493	487.35	<.0001

Panel B

	Split LMT	Regular LMT	P- value (t-test)
Distance from Best Quote All LMT (# of ticks)	-5.69	-11.97	<.0001
Distance from Best Quote Price Improving LMT (# of ticks)	1.78	2.05	<.0001

Table 15: Price Aggressiveness (Market / Limit) and Order Splitting

Table 15 reports the results of a bi-variate probit model for the probability to split a market order and the probability to submit a market order. Each split order is a sequence of orders for the same stock, of the same size and price, submitted on the same side (buy/sell) within an interval of one minute. For the estimation of this model we consider only market order (split / non-split) and non-split limit orders. We measure the Marginal Effect (ME) of a continuous variable X as the effect of one standard deviations from the sample mean on the probability, setting all other variables to sample mean values. For a binary variable D, the calculations is applied to the difference between levels 1 and 0 of D, while all other variables are set to sample mean values. The sample consists of the 40 largest stocks of TASE and the sample period is July to December 2006 (127 trading days). Variable descriptions are provided in Table 1.

Variables	(1) Prob. of Splitting			(2) Prob. of a Market Order	
	Estimate	Marginal Effect*100	Marginal Effect*100 MKT	Estimate	Marginal Effect*100
<i>Intraday & weekly fixed effects</i> [†]					
09:45-11:15	-4.682***	-2.921	-8.376	-0.057***	-2.066
11:15-12:45	-4.583***	-2.246	-6.517	-0.015	-0.552
12:45-14:15	-4.507***	-1.478	-4.291	-0.014	-0.495
14:15-15:45	-4.509***	-1.485	-4.291	-0.041***	-1.490
First day	-1.908***	-0.514	-1.452	-0.099***	-3.553
Mid-week	-2.052***	-4.609	-12.368	-0.103***	-3.770
<i>Market conditions</i>					
Same Side depth	-0.013***	-0.011	-0.065	0.048***	2.625
Opposite Side depth	0.032***	0.027	0.072	0.013***	0.724
Order Arrival Rate	0.010 **	0.009	0.102	-0.096***	-5.597
Trade Volume	0.014***	0.020	0.026	0.021***	1.968
Lag Order Size	-0.045***	-0.025	-0.106	0.061***	2.267
IVolatility	0.389***	0.009	0.011	0.886***	1.391
Spread	-0.077***	-0.052	0.044	-0.343***	-14.712
IVolatility×Spread	-1.081 **	-0.048	-0.118	-0.255 **	-1.006
<i>Order characteristics</i>					
Order Size	0.387***	0.333	0.994	-0.022***	-0.968
Ordersize×last hour	-0.412***	-7.095	-19.339	-0.003 **	-0.456
Ordersize×last day	-0.203***	-0.941	-2.591	-0.008***	-1.288
Buy	0.082***	0.045	-0.150	-0.034***	-1.236
Round	-0.022***	-0.012	-0.941	-0.014***	-0.528
<i>Daily & Cross sectional variables</i>					
Price	0.000 *	0.000	0.000	0.000***	0.007
Volume	-0.229***	-0.153	-0.366	-0.093***	-3.784
ZVolume	-0.000	-0.001	-0.006	-0.000	-0.007
Volatility	-0.902***	-0.028	-0.043	0.695***	1.444
TA25	0.648***	0.214	0.641	0.110***	3.951
Dual Listing	0.021 **	0.012	0.109	0.105***	3.876
Tick Size (0.001) [†]	-0.197***	-0.086	-0.941	0.414***	15.910
Tick Size (0.010) [†]	-0.115***	-0.065	-0.941	0.215***	7.783
Corr(split*, price*)	0.860***				
# Obs	4,263,670		Log Likelihood		-2,705,109

* significant at 10%, ** significant at 5%, *** significant at 1%

[†]Base caeses for intraday and weekly fixed effects are the last one and half hour of the day and the last day of the week respectively. Base case for tick size is 0.1.

Table 16: Splitting of Market and Limit Orders

Table 16 reports the results of a bi-variate probit model for the probability to split a market order and the probability to split a limit order. Each split order is a sequence of orders for the same stock, of the same size and price, submitted on the same side (buy/sell) within an interval of one minute. We measure the Marginal Effect (ME) of a continuous variable X as the effect of one standard deviations from the sample mean on the probability, setting all other variables to sample mean values. For a binary variable D, the calculations is applied to the difference between levels 1 and 0 of D, while all other variables are set to sample mean values. The sample consists of the 40 largest stocks of TASE and the sample period is July to December 2006 (127 trading days). Variable descriptions are provided in Table 1.

Variables	(1) Prob. of Split MKT		(2) Prob. of Split LMT	
	Estimate	Marginal Effect*100	Estimate	Marginal Effect*100
<i>Intraday & weekly fixed effects</i> ^Y				
09:45-11:15	-5.120***	-3.393	-6.001***	-3.618
11:15-12:45	-5.028***	-2.582	-5.921***	-2.668
12:45-14:15	-4.947***	-1.647	-5.880***	-1.607
14:15-15:45	-4.941***	-1.650	-5.889***	-1.621
First day	-2.154 ***	-0.539	-2.730***	-0.458
Mid-week	-2.297***	-5.875	-2.608***	-6.154
<i>Market conditions</i>				
Same Side depth	-0.018***	-0.014	-0.029***	-0.013
Opposite Side depth	0.060***	0.047	-0.017***	-0.008
Order Arrival Rate	-0.004	-0.003	0.209***	0.119
Trade Volume	0.018***	0.023	-0.033***	-0.025
Lag Order Size	-0.037***	-0.019	-0.198***	-0.065
IVolatility	0.550***	0.012	-1.525***	-0.020
Spread	-0.073***	-0.044	0.082***	0.030
IVolatility×Spread	-1.312***	-0.050	-0.769	-0.020
<i>Order characteristics</i>				
Order Size	0.425***	0.352	0.504***	0.296
Ordersize×last hour	-0.453***	-8.993	-0.526***	-11.292
Ordersize×last day	-0.227***	-1.114	-0.255***	-1.002
Buy	0.091***	0.045	0.152***	0.046
Round	-0.029***	-0.014	-0.018***	-0.005
<i>Daily & Cross sectional variables</i>				
Price	0.000	0.000	0.000	0.000
Volume	-0.237***	-0.146	-0.205***	-0.075
ZVolume	-0.000	-0.002	-0.025***	-0.075
Volatility	-0.558***	-0.016	-0.681***	-0.012
TA25	0.666***	0.199	0.308***	0.070
Dual Listing	0.007	0.004	0.117***	0.039
Tick Size (0.001) ^Y	-0.160***	-0.067	-0.168***	-0.042
Tick Size (0.010) ^Y	-0.108***	-0.056	-0.172***	-0.055
Corr(SlitMKT, SplitLMT)	-0.395***			
# Obs	4,279,788		Log Likelihood	
			-186,404	

* significant at 10%, ** significant at 5%, *** significant at 1%

^YBase caeses for intraday and weekly fixed effects are the last one and half hour of the day and the last day of the week respectively. Base case for tick size is 0.1.